

# Chapter 2: Kinematics

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In this Chapter, we will study the motion of particles in space without considering the reasons for the motion; that is what *kinematics* is about. We will introduce the concepts of position, displacement, velocity, and acceleration vectors, and explore their relationships. We will also discuss different types of motion, including linear, circular, spiral, oscillatory, and cycloidal motion. Finally, we will work through some examples to illustrate these concepts. We will stick to Cartesian coordinates for simplicity, but the concepts can be extended to other coordinate systems as well.

## 1 Notion of a Particle

A *particle* is a body whose size and shape are not important for the problem at hand. In other words, a particle can be treated as a point mass. Its motion is described by its position, velocity, and acceleration as functions of time. Mathematically, a particle is a point in space with an associated mass.

For example, planets can be treated as particles when studying their motion around the Sun, even though they have finite size and shape. However, when studying phenomena such as tidal waves or atmospheric motion, the size and shape of the Earth are important, so the Earth cannot be treated as a particle.

*Remark.* Since a particle's entire mass is idealized as being concentrated at a single point, its mass density is formally singular at that point. This is an idealization and requires some mathematical care; a rigorous treatment uses the Dirac delta distribution, which will be introduced in the Electricity and Magnetism course.

## 2 The Position Vector

The position of a particle is described by a vector called the *position vector*:

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}, \quad (1)$$

where  $x(t)$ ,  $y(t)$ , and  $z(t)$  are the Cartesian coordinates of the particle as functions of time, and  $\mathbf{i}$ ,  $\mathbf{j}$ , and  $\mathbf{k}$  are the unit vectors in the  $x$ -,  $y$ -, and  $z$ -directions, respectively.

The position vector points from the origin of the coordinate system to the particle. Its magnitude,

$$|\mathbf{r}(t)| = \sqrt{x^2(t) + y^2(t) + z^2(t)}, \quad (2)$$

is the distance of the particle from the origin, while its direction specifies the direction from the origin to the particle.

Because the coordinates can change with time, the position vector  $\mathbf{r}(t)$  describes the motion of the particle through space.

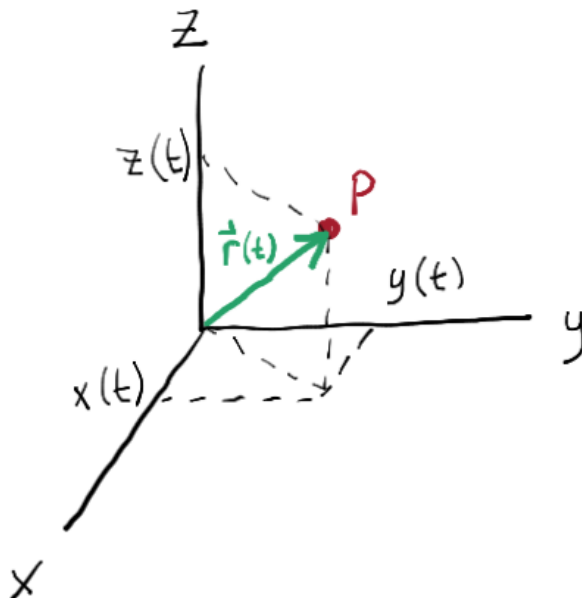


Figure 1: The position vector  $\mathbf{r}(t)$  of a particle in a three-dimensional Cartesian coordinate system.

### 3 The Displacement Vector

Suppose a particle moves from an initial position  $\mathbf{r}(t_1)$  to a final position  $\mathbf{r}(t_2)$ . The *displacement vector* describes the change in the particle's position. It is defined as the difference between the final and initial position vectors:

$$\Delta \mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1). \quad (3)$$

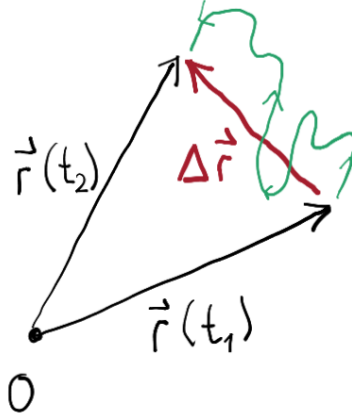


Figure 2: Displacement vector of a particle moving from an initial position  $\mathbf{r}(t_1)$  to a final position  $\mathbf{r}(t_2)$ . The actual path, which is shown by the green curve, taken by the particle is not necessarily a straight line.

The displacement vector points from the initial position to the final position. Its magnitude,  $|\Delta \mathbf{r}|$ , is the straight-line distance between the initial and final positions. In general, it is not equal to the distance traveled by the particle along its trajectory. The displacement depends only on the initial and final positions, not on the path taken between them.

Geometrically, the relation given in Eq. (3) follows directly from vector subtraction. Since both position vectors originate at the origin, the vector connecting the initial position  $P_1$  to the final position  $P_2$  is equal to  $\mathbf{r}(t_2) - \mathbf{r}(t_1)$ . Thus, the displacement vector is the vector from the initial position to the final position.

## 4 The Average Velocity Vector

Velocity describes how the position of a particle changes with time. Suppose the particle moves from an initial position  $\mathbf{r}(t_1)$  to a final position  $\mathbf{r}(t_2)$  during the time interval  $\Delta t = t_2 - t_1$  and let the corresponding displacement be  $\Delta \mathbf{r}$ . We define *the average velocity vector* as

$$\mathbf{v}_{\text{avg}} = \frac{\Delta \mathbf{r}}{\Delta t}. \quad (4)$$

This quantity describes the displacement per unit time during the interval  $\Delta t$ . Note that the particle may have sped up, slowed down, or even stopped temporarily during the motion from  $\mathbf{r}(t_1)$  to  $\mathbf{r}(t_2)$ . From the initial and final positions alone, we can therefore determine only the particle's motion *on average* over the time interval as given in Eq. (4). In the way the displacement shows where the particle went on average, the average velocity vector shows the direction and magnitude of the particle's motion on average. Consequently, the average velocity vector is in the same direction as the displacement vector, provided  $\Delta \mathbf{r} \neq \mathbf{0}$ .

Given that the initial and final position vectors be

$$\mathbf{r}(t_1) = x(t_1)\mathbf{i} + y(t_1)\mathbf{j} + z(t_1)\mathbf{k}, \quad (5)$$

$$\mathbf{r}(t_2) = x(t_2)\mathbf{i} + y(t_2)\mathbf{j} + z(t_2)\mathbf{k}, \quad (6)$$

the displacement can be written as

$$\begin{aligned} \Delta\mathbf{r} &= \mathbf{r}(t_2) - \mathbf{r}(t_1) \\ &= [x(t_2) - x(t_1)]\mathbf{i} + [y(t_2) - y(t_1)]\mathbf{j} + [z(t_2) - z(t_1)]\mathbf{k}. \end{aligned} \quad (7)$$

Defining

$$\Delta x = x(t_2) - x(t_1), \quad \Delta y = y(t_2) - y(t_1), \quad \Delta z = z(t_2) - z(t_1), \quad (8)$$

the average velocity can also be written as

$$\mathbf{v}_{\text{avg}} = \frac{\Delta x}{\Delta t}\mathbf{i} + \frac{\Delta y}{\Delta t}\mathbf{j} + \frac{\Delta z}{\Delta t}\mathbf{k}. \quad (9)$$

## 5 The Instantaneous Velocity Vector

To determine the velocity at a particular instant, we consider the average velocity over an increasingly small time interval around that instant. Mathematically, considering the limit of the average velocity as  $\Delta t \rightarrow 0$  leads us to the definition of *instantaneous velocity*

$$\mathbf{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta\mathbf{r}}{\Delta t} = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}. \quad (10)$$

Using Newton's notation,

$$\mathbf{v}(t) = \dot{x}(t)\mathbf{i} + \dot{y}(t)\mathbf{j} + \dot{z}(t)\mathbf{k} = v_x(t)\mathbf{i} + v_y(t)\mathbf{j} + v_z(t)\mathbf{k}. \quad (11)$$

The *speed* of the particle is the magnitude of its velocity vector:

$$v(t) = |\mathbf{v}(t)|. \quad (12)$$

We use speed if we are only interested in how fast the particle is moving, regardless of its direction of motion. The speed is always a non-negative quantity, while the velocity can be positive or negative depending on the direction of motion.

*Remark.* Compared to the average velocity, calculating the instantaneous velocity requires more detailed information about the motion of the particle. In calculating displacement, we only needed two vectors, the initial and final positions. However, in calculating the instantaneous velocity, we need to know  $\mathbf{r}(t)$ , i.e., all of the position vectors at each instant in time.

## 6 The Average Acceleration Vector

Acceleration is defined as the change in velocity with respect to time. Following the same reasoning as we did for the average velocity in Sec. 4, we can define the average acceleration vector as follows:

$$\mathbf{a}_{\text{avg}} = \frac{\Delta \mathbf{v}}{\Delta t} \quad (13)$$

where  $\Delta \mathbf{v} = \mathbf{v}_2 - \mathbf{v}_1$  is the change in velocity vector, and  $\Delta t = t_2 - t_1$  is the time interval over which the change in velocity occurs. The average acceleration vector describes how the velocity of the particle changes on average over the time interval  $\Delta t$ . The average acceleration vector has the same direction as the change in velocity vector. If we are given the initial and final velocity vectors in component form as

$$\mathbf{v}(t_1) = v_x(t_1)\mathbf{i} + v_y(t_1)\mathbf{j} + v_z(t_1)\mathbf{k}$$

$$\mathbf{v}(t_2) = v_x(t_2)\mathbf{i} + v_y(t_2)\mathbf{j} + v_z(t_2)\mathbf{k}$$

Then the change in velocity is

$$\Delta \mathbf{v} = \mathbf{v}(t_2) - \mathbf{v}(t_1) = [v_x(t_2) - v_x(t_1)]\mathbf{i} + [v_y(t_2) - v_y(t_1)]\mathbf{j} + [v_z(t_2) - v_z(t_1)]\mathbf{k}$$

and the average acceleration vector becomes

$$\mathbf{a}_{\text{avg}} = \frac{\Delta v_x}{\Delta t}\mathbf{i} + \frac{\Delta v_y}{\Delta t}\mathbf{j} + \frac{\Delta v_z}{\Delta t}\mathbf{k}$$

## 7 The Instantaneous Acceleration Vector

If we are interested in the acceleration of the particle at a given time instead of the average acceleration we should consider the behavior of  $\mathbf{a}_{\text{avg}}$  when  $\Delta t \rightarrow 0$ . Therefore,

$$\lim_{\Delta t \rightarrow 0} \mathbf{a}_{\text{avg}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t} = \frac{d\mathbf{v}}{dt} = \frac{dv_x}{dt}\mathbf{i} + \frac{dv_y}{dt}\mathbf{j} + \frac{dv_z}{dt}\mathbf{k}$$

where we have the definition of derivative for each term. Hence the acceleration of a particle at time  $t$  is defined as

$$\mathbf{a}(t) = \frac{d}{dt}[v_x(t)]\mathbf{i} + \frac{d}{dt}[v_y(t)]\mathbf{j} + \frac{d}{dt}[v_z(t)]\mathbf{k}. \quad (14)$$

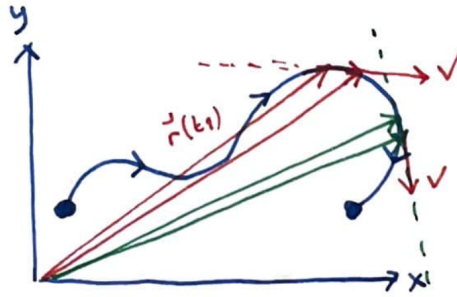
or in Newton's notation

$$\mathbf{a}(t) = \dot{v}_x(t)\mathbf{i} + \dot{v}_y(t)\mathbf{j} + \dot{v}_z(t)\mathbf{k} = a_x(t)\mathbf{i} + a_y(t)\mathbf{j} + a_z(t)\mathbf{k}. \quad (15)$$

where  $a_x(t)$ ,  $a_y(t)$ , and  $a_z(t)$  are the components of the acceleration vector in the  $x$ ,  $y$ , and  $z$  directions, respectively.

## 8 The Trajectory of a Particle

The *trajectory* of a particle is the path followed by the particle in space as time goes on. The *trajectory equation* of a particle is the equation that describes the trajectory of the particle in space. It can be obtained by eliminating the time variable from the position vector equation  $\mathbf{r}(t)$  and connecting the  $x$ ,  $y$ , and  $z$  coordinates of the particle.



## 9 Connection between position, velocity, and acceleration vectors

We have already seen that once we are given the position vector of a particle,  $\mathbf{r}(t)$ , we can find the velocity and acceleration vectors by taking the first and second derivatives with respect to time, respectively:

$$\mathbf{r}(t) \xrightarrow{\frac{d}{dt}} \mathbf{v}(t) \xrightarrow{\frac{d}{dt}} \mathbf{a}(t) \quad (16)$$

Is it possible to go the other way around? If we are given the acceleration vector of a particle,  $\mathbf{a}(t)$ , can we find the velocity and position vectors? If we are given the velocity vector, for example, can we find  $\mathbf{r}(t)$ ?



The area under the velocity-time graph represents the displacement of the particle. Therefore, we can use integral to calculate the displacement of the particle between two time instants  $t_1$  and  $t_2$ :

$$\Delta x = \int_{t_1}^{t_2} v_x(t) dt \quad (17)$$

However, we need to know the initial position of the particle at time  $t_1$  in order to find the final position of the particle at time  $t_2$ . Therefore, we can write:

$$x(t_2) = x(t_1) + \int_{t_1}^{t_2} v_x(t) dt \quad (18)$$

Therefore, if we are only given the form of  $\mathbf{v}(t)$  but not the initial position of the particle, we can find the form of  $\mathbf{r}(t)$  up to an unknown constant  $\mathbf{C}$ , which actually comes automatically with the integration of the indefinite integral:

$$\mathbf{r}(t) = \int \mathbf{v}(t) dt + \mathbf{C} \quad (19)$$

where  $\mathbf{C}$  is a constant vector that represents the initial position of the particle at time  $t = 0$ . The components of  $\mathbf{C}$  can be determined if we are given the initial position of the particle. Like derivative, the integral is also a linear operator, which means that we can integrate each component of the velocity vector separately to find the corresponding components of the position vector:

$$\mathbf{r}(t) = \int v_x(t) dt \mathbf{i} + \int v_y(t) dt \mathbf{j} + \int v_z(t) dt \mathbf{k} + \mathbf{C} \quad (20)$$

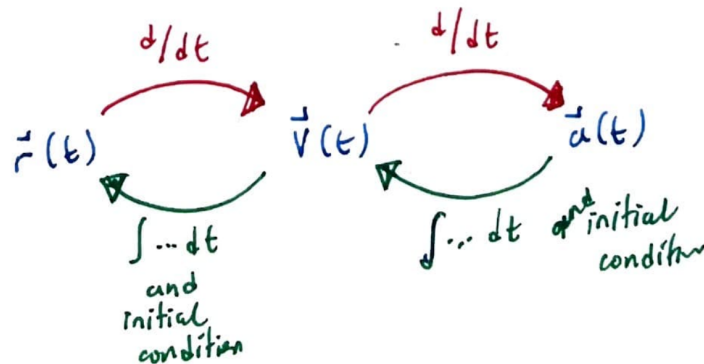
The velocity is the time rate of change of the displacement and we saw that we can derive the position out of the velocity by using integration. Also, acceleration is the time rate of change of velocity, which means that the mathematical derivation of the velocity out of the acceleration have to be in the same way we derived the position out of the velocity:

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt + \mathbf{C} \quad (21)$$

## 10 Summary

If we are given how the position vector of a particle is changing with time, it means that we know everything about the motion of the particle. We can find the velocity and acceleration vectors by taking the first and second derivatives of the position vector with respect to time, respectively. If we are given how the acceleration vector of a particle is changing with time, we can find the velocity and position vectors by taking the indefinite integrals of the acceleration vector with respect to time, respectively. However, we need to know the initial conditions of the particle in order to find its exact position and velocity at any given time.

(Note for myself: Put some more text here to explain my information point of view.)



## 11 Types of Motion

### 11.1 Linear Motion

Motion along a straight line is called linear motion. In this case, the position vector of the particle can be expressed as:

$$\mathbf{r}(t) = x(t)\mathbf{i}$$

Here,  $x$  direction is chosen randomly as the direction of motion. The other two directions could also be chosen equally well. The velocity and acceleration vectors are also along the same direction as the position vector, and can be expressed as:

$$\mathbf{v}(t) = \dot{x}(t)\mathbf{i}$$

$$\mathbf{a}(t) = \ddot{x}(t)\mathbf{i}$$

When the motion is restricted to a straight line, or in other words, when the motion is one-dimensional, the position, velocity, and acceleration vectors can be represented by scalar quantities along the direction of motion:

$$x = x(t), v = \dot{x}(t), a = \ddot{x}(t)$$

The signs of these scalar quantities indicate the direction of motion along the chosen axis. For example, if the particle is moving in the positive  $x$  direction, the position, velocity, and acceleration scalars will be positive. If the particle is moving in the negative  $x$  direction, these scalars will be negative. The magnitude of these scalars represents the distance traveled, speed, and rate of change of speed, respectively.

## 11.2 Circular Motion

*Circular motion* is the motion of a particle along a circular path:

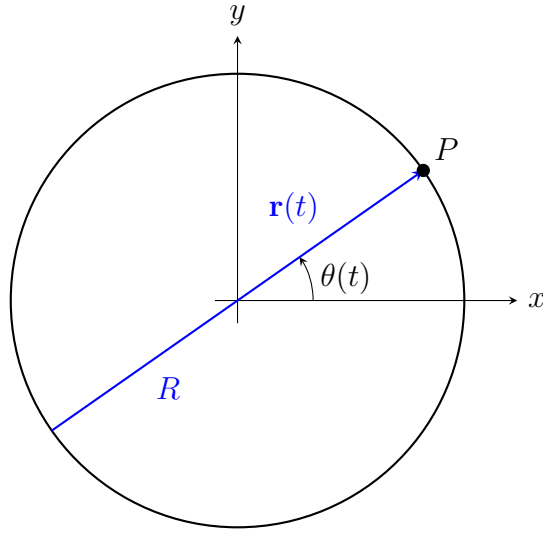


Figure 3: Circular path and polar-coordinate vectors at the particle's position.

$$\mathbf{r}(t) = R \cos[\theta(t)]\mathbf{i} + R \sin[\theta(t)]\mathbf{j}$$

Here,  $R$  is the radius of the circular path, and  $\theta(t)$  is the angle made by the position vector with respect to a reference axis. We know that  $\theta$  changes with time, so it can be represented as a function of time:

$$\theta = \theta(t)$$

There is no mathematical restriction on the form of  $\theta(t)$ , so it can be any function of time. However, if we are interested in *uniform circular motion*, where the speed of the particle is constant, the angle  $\theta$  is defined as a linear function of time:

$$\theta(t) = \omega t + \theta_0$$

Here  $\omega$  is the *angular velocity* of the particle, and  $\theta_0$  is the initial angle of the particle at time  $t = 0$ . The angular velocity  $\omega$  is defined as the rate of change of the angle with respect to time:

$$\omega = \frac{d\theta}{dt}$$

**What is 1 radian?** A radian is a unit of angular measure used in mathematics and physics. It is defined as the angle subtended at the center of a circle by an arc whose length is equal to the radius of the circle. In other words, if you take a circle and draw an arc that has the same length as the radius of the circle, the angle formed at the center of the circle by that arc is 1 radian.

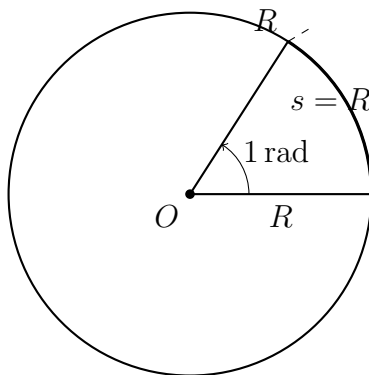
If 1 radian sees the arc length  $R$  from the center of the circle,  $2\pi$  radians sees the whole circumference of the circle, which is  $2\pi R$ . Therefore, we can write:

$$2\pi \text{ radians} = 360^\circ$$

Also, if  $R$  is seen by 1 radian, any other angle  $\theta$  sees the arc length  $s$  as follows:

$$\theta = \frac{s}{R}$$

We will be using definition of radian in the following sections, so it is important to understand it well. The following figure shows the definition of one radian:



Arc length  $s$  equals the radius  $R$

$$\theta = \frac{s}{R} = 1 \text{ rad}$$

Figure 4: Definition of one radian: an angle of 1 radian subtends an arc whose length equals the radius of the circle.

What about the velocity vector of a particle moving along a circular path? We can find the velocity vector by taking the derivative of the position vector with respect to time:

$$\mathbf{v}(t) = \frac{d}{dt}[R \cos(\omega t)]\mathbf{i} + \frac{d}{dt}[R \sin(\omega t)]\mathbf{j} \quad (22)$$

$$= -R \sin(\omega t)\omega \mathbf{i} + R \cos(\omega t)\omega \mathbf{j} \quad (23)$$

### 11.3 Spiral Motion

It is a composite motion of linear and circular motions. The particle moves along a spiral path, which can be described by a combination of linear and circular motion equations. The

position vector of a particle moving in a spiral path can be expressed as:

$$\mathbf{r}(t) = R \cos[\theta(t)]\mathbf{i} + R \sin[\theta(t)]\mathbf{j} + z(t)\mathbf{k}$$

## 11.4 Oscillatory Motion

The exponential function  $e^x$  is often recognized for its unique property that its derivative is equal to itself. It is an analytical function that can be defined through its Taylor series expansion:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad (24)$$

Couldn't we use a cosine or sine function to describe oscillations instead of complex numbers? The answer is yes, but complex numbers provide a more elegant and efficient way to handle oscillatory phenomena, especially when dealing with multiple oscillations or when performing mathematical operations such as differentiation and integration. Using Euler's formula, we can express the complex exponential function as:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (25)$$

This relationship allows us to rewrite the polar form of a complex number in terms of the exponential function:

$$z(t) = r e^{i(\omega t + \phi)} \quad (26)$$

This representation is particularly useful in physics and engineering, as it simplifies the analysis of oscillatory systems. For example, when dealing with differential equations that describe oscillatory motion, using complex exponentials can make the calculations more straightforward. Additionally, complex numbers facilitate the representation of phase relationships between different oscillations, which is essential in many applications such as signal processing and wave mechanics.

## 11.5 Cycloidal Motion — *Optional*

It is the path traced by a point on the circumference of a circle as it rolls along straight line without slipping. Imagine drawing a chalk dot on a bicycle wheel and then rolling the wheel along a flat surface. The path traced by the chalk dot is a cycloid.

The position vector of a particle moving along a cycloidal path can be expressed as:

$$\mathbf{r}(t) = R(t - \sin t)\mathbf{i} + R(1 - \cos t)\mathbf{j}$$

## 12 Worked Examples

### 12.1 Collision of Two Particles

Two particles are moving on a plane according to the following position vectors:

$$\mathbf{r}_1(t) = t\mathbf{i} + t\mathbf{j}$$

$$\mathbf{r}_2(t) = (-t + 4)\mathbf{i} + (2t - 2)\mathbf{j}$$

What time  $t$  do the two particles collide? What is the position vector of the collision point?

**Solution 1:** The two particles collide when their position vectors are equal:

$$\mathbf{r}_1(t) = \mathbf{r}_2(t)$$

This gives us two equations:

$$t = -t + 4 \quad (\text{i})$$

$$t = 2t - 2 \quad (\text{ii})$$

From equation (i), we get:

$$2t = 4 \implies t = 2$$

From equation (ii), we get:

$$t = 2t - 2 \implies t = 2$$

Both equations give us the same time  $t = 2$ . Now we can find the position vector of the collision point by substituting  $t = 2$  into either  $\mathbf{r}_1(t)$  or  $\mathbf{r}_2(t)$ :

$$\mathbf{r}_1(2) = 2\mathbf{i} + 2\mathbf{j} = 2\mathbf{i} + 2\mathbf{j}$$

**Solution 2:** Define a *separation vector*  $\Delta\mathbf{r}$ :

$$\Delta\mathbf{r} = \mathbf{r}_2(t) - \mathbf{r}_1(t)$$

The two particles collide when  $|\Delta\mathbf{r}| = 0$ :

$$|(-2t + 4)\mathbf{i} + (t - 2)\mathbf{j}| = 0$$

This is equivalent to:

$$\sqrt{(-2t + 4)^2 + (t - 2)^2} = 0$$

Squaring both sides:

$$(-2t + 4)^2 + (t - 2)^2 = 0$$

Expanding:

$$4t^2 - 16t + 16 + t^2 - 4t + 4 = 0$$

Combining like terms:

$$5t^2 - 20t + 20 = 0$$

Dividing by 5:

$$t^2 - 4t + 4 = 0$$

This factors as:

$$(t - 2)^2 = 0$$

So  $t = 2$ .

## 13 Worked Examples

### 13.1 Displacement, Average Velocity, Instantaneous Velocity, and Collision

Two particles move in the  $xy$ -plane according to

$$\mathbf{r}_1(t) = (t^2 + 1)\mathbf{i} + 2t\mathbf{j},$$

$$\mathbf{r}_2(t) = (5 - t)\mathbf{i} + (t^2 - 1)\mathbf{j}.$$

1. Find the position of each particle at  $t = 0$  and  $t = 2$ .
2. Find the displacement vectors of both particles between  $t = 0$  and  $t = 2$ .
3. Find their average velocity vectors over this time interval.
4. Find  $\mathbf{v}_1(t)$  and  $\mathbf{v}_2(t)$ .
5. Determine whether the particles collide. If they do, find the collision time and collision position.

**Solution:**

**(a) Positions at  $t = 0$  and  $t = 2$**

For particle 1,

$$\mathbf{r}_1(0) = (0^2 + 1)\mathbf{i} + 2(0)\mathbf{j} = \mathbf{i},$$

and

$$\mathbf{r}_1(2) = (2^2 + 1)\mathbf{i} + 2(2)\mathbf{j} = 5\mathbf{i} + 4\mathbf{j}.$$

For particle 2,

$$\mathbf{r}_2(0) = (5 - 0)\mathbf{i} + (0^2 - 1)\mathbf{j} = 5\mathbf{i} - \mathbf{j},$$

and

$$\mathbf{r}_2(2) = (5 - 2)\mathbf{i} + (2^2 - 1)\mathbf{j} = 3\mathbf{i} + 3\mathbf{j}.$$

Therefore,

$$\boxed{\mathbf{r}_1(0) = \mathbf{i}, \quad \mathbf{r}_1(2) = 5\mathbf{i} + 4\mathbf{j}}$$

and

$$\boxed{\mathbf{r}_2(0) = 5\mathbf{i} - \mathbf{j}, \quad \mathbf{r}_2(2) = 3\mathbf{i} + 3\mathbf{j}.$$

**(b) Displacement vectors**

The displacement is

$$\Delta \mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1).$$

Thus, for particle 1,

$$\Delta \mathbf{r}_1 = \mathbf{r}_1(2) - \mathbf{r}_1(0) = (5\mathbf{i} + 4\mathbf{j}) - \mathbf{i},$$

so

$$\boxed{\Delta \mathbf{r}_1 = 4\mathbf{i} + 4\mathbf{j}.$$

For particle 2,

$$\Delta \mathbf{r}_2 = \mathbf{r}_2(2) - \mathbf{r}_2(0) = (3\mathbf{i} + 3\mathbf{j}) - (5\mathbf{i} - \mathbf{j}),$$

hence

$$\boxed{\Delta \mathbf{r}_2 = -2\mathbf{i} + 4\mathbf{j}.$$

**(c) Average velocity vectors**

The average velocity is

$$\mathbf{v}_{\text{avg}} = \frac{\Delta \mathbf{r}}{\Delta t}.$$

Since

$$\Delta t = 2 - 0 = 2,$$

we obtain

$$\mathbf{v}_{\text{avg},1} = \frac{4\mathbf{i} + 4\mathbf{j}}{2} = 2\mathbf{i} + 2\mathbf{j},$$

and therefore

$$\boxed{\mathbf{v}_{\text{avg},1} = 2\mathbf{i} + 2\mathbf{j}.}$$

Similarly,

$$\mathbf{v}_{\text{avg},2} = \frac{-2\mathbf{i} + 4\mathbf{j}}{2} = -\mathbf{i} + 2\mathbf{j},$$

so

$$\boxed{\mathbf{v}_{\text{avg},2} = -\mathbf{i} + 2\mathbf{j}.}$$

#### (d) Instantaneous velocity

The instantaneous velocity is obtained from

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

For particle 1,

$$\mathbf{v}_1(t) = \frac{d}{dt} [(t^2 + 1)\mathbf{i} + 2t\mathbf{j}] ,$$

giving

$$\boxed{\mathbf{v}_1(t) = 2t\mathbf{i} + 2\mathbf{j}.}$$

For particle 2,

$$\mathbf{v}_2(t) = \frac{d}{dt} [(5 - t)\mathbf{i} + (t^2 - 1)\mathbf{j}] ,$$

so

$$\boxed{\mathbf{v}_2(t) = -\mathbf{i} + 2t\mathbf{j}.}$$

**(e) Collision**

The particles collide when their position vectors are equal at the same time:

$$\mathbf{r}_1(t) = \mathbf{r}_2(t).$$

This gives the two equations

$$t^2 + 1 = 5 - t,$$

and

$$2t = t^2 - 1.$$

The first equation gives

$$t^2 + t - 4 = 0,$$

so

$$t = \frac{-1 \pm \sqrt{17}}{2}.$$

The second equation gives

$$t^2 - 2t - 1 = 0,$$

so

$$t = 1 \pm \sqrt{2}.$$

There is no time that satisfies both equations simultaneously. Therefore, the particles do not collide.

The two particles never collide.

This illustrates that equality of one coordinate is not sufficient for a collision; all spatial coordinates must be equal at the same instant.

## 13.2 Velocity, Acceleration, and the Interpretation of Signs

A particle moves along the  $x$ -axis according to

$$x(t) = t^3 - 6t^2 + 9t.$$

1. Find  $v(t)$  and  $a(t)$ .
2. Determine the times at which the particle is at rest.
3. For each time interval separated by these instants, determine whether the particle is moving in the  $+x$  or  $-x$  direction.
4. Determine where the particle is speeding up and where it is slowing down.
5. Find the total distance traveled during  $0 \leq t \leq 4$ .
6. Compare the total distance traveled with the magnitude of the net displacement.

**Solution:**

**(a) Velocity and acceleration**

The velocity is the derivative of position:

$$v(t) = \frac{dx}{dt} = 3t^2 - 12t + 9.$$

Factoring,

$$v(t) = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3).$$

Therefore,

$$\boxed{v(t) = 3(t - 1)(t - 3).}$$

The acceleration is

$$a(t) = \frac{dv}{dt} = 6t - 12,$$

so

$$\boxed{a(t) = 6(t - 2).}$$

**(b) Times at which the particle is at rest**

The particle is at rest when

$$v(t) = 0.$$

Therefore,

$$3(t - 1)(t - 3) = 0,$$

which gives

$$\boxed{t = 1, \quad t = 3.}$$

**(c) Direction of motion**

The sign of  $v(t)$  determines the direction of motion.

For  $0 < t < 1$ ,

$$(t - 1) < 0, \quad (t - 3) < 0,$$

so

$$v(t) > 0.$$

Thus the particle moves in the  $+x$  direction.

For  $1 < t < 3$ ,

$$(t - 1) > 0, \quad (t - 3) < 0,$$

so

$$v(t) < 0.$$

Thus the particle moves in the  $-x$  direction.

For  $t > 3$ ,

$$(t - 1) > 0, \quad (t - 3) > 0,$$

so

$$v(t) > 0.$$

Thus the particle again moves in the  $+x$  direction.

Therefore,

| Time interval | Direction |
|---------------|-----------|
| $0 < t < 1$   | $+x$      |
| $1 < t < 3$   | $-x$      |
| $t > 3$       | $+x$      |

#### (d) Speeding up and slowing down

A particle speeds up when the velocity and acceleration have the same sign, and slows down when they have opposite signs.

We have

$$a(t) = 6(t - 2).$$

Hence,

$$a(t) < 0 \quad \text{for } t < 2,$$

and

$$a(t) > 0 \quad \text{for } t > 2.$$

Combining this with the sign of  $v(t)$ :

| Interval    | $v$ | $a$ | Behavior     |
|-------------|-----|-----|--------------|
| $0 < t < 1$ | +   | -   | slowing down |
| $1 < t < 2$ | -   | -   | speeding up  |
| $2 < t < 3$ | -   | +   | slowing down |
| $t > 3$     | +   | +   | speeding up  |

Thus,

|                                                                                       |
|---------------------------------------------------------------------------------------|
| slowing down: $0 < t < 1, \quad 2 < t < 3,$<br>speeding up: $1 < t < 2, \quad t > 3.$ |
|---------------------------------------------------------------------------------------|

**(e) Total distance traveled for  $0 \leq t \leq 4$**

Because the particle changes direction at  $t = 1$  and  $t = 3$ , we must divide the motion into separate intervals.

First calculate the positions:

$$x(0) = 0,$$

$$x(1) = 1 - 6 + 9 = 4,$$

$$x(3) = 27 - 54 + 27 = 0,$$

$$x(4) = 64 - 96 + 36 = 4.$$

Therefore, the distances traveled are

$$d_1 = |x(1) - x(0)| = |4 - 0| = 4,$$

$$d_2 = |x(3) - x(1)| = |0 - 4| = 4,$$

$$d_3 = |x(4) - x(3)| = |4 - 0| = 4.$$

Hence the total distance is

$$\boxed{d_{\text{total}} = 4 + 4 + 4 = 12.}$$

**(f) Distance versus displacement**

The net displacement is

$$\Delta x = x(4) - x(0) = 4 - 0 = 4.$$

Its magnitude is therefore

$$|\Delta x| = 4.$$

Thus,

|                                               |
|-----------------------------------------------|
| total distance = 12,       displacement  = 4. |
|-----------------------------------------------|

The difference occurs because the particle changes direction during its motion.

### 13.3 Reconstructing Position and Velocity from Acceleration

A particle moves in one dimension with acceleration

$$a(t) = 6t - 4.$$

At  $t = 0$ ,

$$x(0) = 2, \quad v(0) = -3.$$

1. Find  $v(t)$ .
2. Find  $x(t)$ .
3. Determine the time at which the particle changes direction.
4. Find the position of the particle at that instant.
5. Verify the result by differentiating  $x(t)$ .

**Solution:**

**(a) Finding the velocity**

Since

$$a(t) = \frac{dv}{dt},$$

we integrate:

$$v(t) = \int (6t - 4) dt.$$

Therefore,

$$v(t) = 3t^2 - 4t + C_1.$$

Using the initial condition  $v(0) = -3$ ,

$$-3 = C_1.$$

Hence,

$$\boxed{v(t) = 3t^2 - 4t - 3.}$$

**(b) Finding the position**

Since

$$v(t) = \frac{dx}{dt},$$

we integrate again:

$$x(t) = \int (3t^2 - 4t - 3) dt.$$

Thus,

$$x(t) = t^3 - 2t^2 - 3t + C_2.$$

Using  $x(0) = 2$ ,

$$2 = C_2.$$

Therefore,

$$\boxed{x(t) = t^3 - 2t^2 - 3t + 2.}$$

This demonstrates why the initial conditions are necessary: one integration introduces one unknown constant, and a second integration introduces another.

**(c) Time at which the particle changes direction**

The particle changes direction when its velocity changes sign. First find when it is at rest:

$$v(t) = 0.$$

Thus,

$$3t^2 - 4t - 3 = 0.$$

Using the quadratic formula,

$$t = \frac{4 \pm \sqrt{16 + 36}}{6} = \frac{4 \pm 2\sqrt{13}}{6}.$$

Therefore,

$$t = \frac{2 \pm \sqrt{13}}{3}.$$

Since physical time in this problem is taken to begin at  $t = 0$ , the relevant solution is

$$t_{\text{turn}} = \frac{2 + \sqrt{13}}{3}.$$

#### **(d) Position at the turning point**

Substituting

$$t_{\text{turn}} = \frac{2 + \sqrt{13}}{3}$$

into

$$x(t) = t^3 - 2t^2 - 3t + 2$$

gives the turning-point position

$$x_{\text{turn}} = -\frac{52 + 26\sqrt{13}}{27}.$$

Numerically,

$$x_{\text{turn}} \approx -5.412.$$

#### **(e) Verification**

Differentiating the position function,

$$x(t) = t^3 - 2t^2 - 3t + 2,$$

gives

$$\frac{dx}{dt} = 3t^2 - 4t - 3 = v(t),$$

and differentiating once more,

$$\frac{d^2x}{dt^2} = 6t - 4 = a(t).$$

Therefore,

$$\boxed{\frac{dx}{dt} = v(t), \quad \frac{d^2x}{dt^2} = a(t),}$$

as required.

## 13.4 Uniform Circular Motion

A particle moves on a circle of radius  $R$  according to

$$\mathbf{r}(t) = R \cos(\omega t) \mathbf{i} + R \sin(\omega t) \mathbf{j}.$$

1. Find  $\mathbf{v}(t)$ .
2. Find the speed  $|\mathbf{v}(t)|$ .
3. Find  $\mathbf{a}(t)$ .
4. Show explicitly that

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = 0.$$

5. Show that the acceleration always points toward the center of the circle.
6. Find the magnitude of the acceleration in terms of  $R$  and  $\omega$ .
7. Explain geometrically why the velocity is tangent to the trajectory while the acceleration is directed toward the center.

**Solution:**

**(a) Velocity**

The velocity is

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

Therefore,

$$\mathbf{v}(t) = -R\omega \sin(\omega t)\mathbf{i} + R\omega \cos(\omega t)\mathbf{j}.$$

Hence,

$$\boxed{\mathbf{v}(t) = -R\omega \sin(\omega t)\mathbf{i} + R\omega \cos(\omega t)\mathbf{j}.}$$

**(b) Speed**

The speed is the magnitude of the velocity:

$$|\mathbf{v}(t)| = \sqrt{R^2\omega^2 \sin^2(\omega t) + R^2\omega^2 \cos^2(\omega t)}.$$

Using

$$\sin^2 \theta + \cos^2 \theta = 1,$$

we obtain

$$|\mathbf{v}(t)| = R|\omega|.$$

For  $\omega > 0$ ,

$$\boxed{v = R\omega.}$$

Thus the speed is constant.

**(c) Acceleration**

Differentiate the velocity:

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt}.$$

Hence,

$$\mathbf{a}(t) = -R\omega^2 \cos(\omega t)\mathbf{i} - R\omega^2 \sin(\omega t)\mathbf{j}.$$

Factorizing,

$$\boxed{\mathbf{a}(t) = -\omega^2 [R \cos(\omega t)\mathbf{i} + R \sin(\omega t)\mathbf{j}].}$$

Since the expression in brackets is precisely  $\mathbf{r}(t)$ ,

$$\boxed{\mathbf{a}(t) = -\omega^2 \mathbf{r}(t).}$$

#### (d) Orthogonality of position and velocity

Using the expressions above,

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = [R \cos(\omega t)\mathbf{i} + R \sin(\omega t)\mathbf{j}] \cdot [-R\omega \sin(\omega t)\mathbf{i} + R\omega \cos(\omega t)\mathbf{j}] \quad (27)$$

$$= -R^2\omega \cos(\omega t) \sin(\omega t) + R^2\omega \sin(\omega t) \cos(\omega t) \quad (28)$$

$$= 0. \quad (29)$$

Therefore,

$$\boxed{\mathbf{r}(t) \perp \mathbf{v}(t).}$$

Since the position vector points from the center to the particle, the velocity is perpendicular to the radius and therefore tangent to the circle.

#### (e) Direction of the acceleration

We found that

$$\mathbf{a}(t) = -\omega^2 \mathbf{r}(t).$$

The factor  $-\omega^2$  is negative, so  $\mathbf{a}(t)$  points in the direction opposite to  $\mathbf{r}(t)$ .

Since  $\mathbf{r}(t)$  points from the center toward the particle, the acceleration points from the particle toward the center.

Thus,

$$\boxed{\text{the acceleration is always directed toward the center.}}$$

### (f) Magnitude of the acceleration

Since

$$\mathbf{a}(t) = -\omega^2 \mathbf{r}(t),$$

we have

$$|\mathbf{a}(t)| = \omega^2 |\mathbf{r}(t)|.$$

Because

$$|\mathbf{r}(t)| = R,$$

it follows that

$$a = R\omega^2.$$

Using  $v = R\omega$ , this can also be written as

$$a = \frac{v^2}{R}.$$

### (g) Geometrical interpretation

The particle remains at a fixed distance  $R$  from the center. Therefore, its position vector continuously rotates.

The velocity is the rate of change of the position vector. An infinitesimal change in the position vector along a circular path is tangent to the circle, so the instantaneous velocity is tangent to the trajectory.

The velocity vector itself changes direction continuously. Its change is directed toward the center of the circle, producing an inward acceleration.

Thus,

$$\mathbf{v} \text{ is tangent to the trajectory,} \quad \mathbf{a} \text{ points toward the center.}$$

## 13.5 Oscillatory Motion and the Complex Exponential

Consider the complex function

$$z(t) = Ae^{i(\omega t + \phi)},$$

where  $A$ ,  $\omega$ , and  $\phi$  are constants.

1. Using Euler's formula, write  $z(t)$  in terms of its real and imaginary parts.
2. What trajectory does the corresponding point follow in the complex plane?
3. Calculate  $\dot{z}(t)$  and  $\ddot{z}(t)$ .
4. Show that

$$\ddot{z} + \omega^2 z = 0.$$

5. What are the real and imaginary parts of this equation?
6. Explain why the complex exponential is useful for describing oscillatory motion.

**Solution:**

**(a) Real and imaginary parts**

Using Euler's formula,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

we obtain

$$z(t) = A [\cos(\omega t + \phi) + i \sin(\omega t + \phi)].$$

Therefore,

$$\boxed{\operatorname{Re}[z(t)] = A \cos(\omega t + \phi)}$$

and

$$\boxed{\operatorname{Im}[z(t)] = A \sin(\omega t + \phi).}$$

Thus, the same complex function contains both sine and cosine oscillations.

**(b) Trajectory in the complex plane**

The real and imaginary coordinates are

$$x(t) = A \cos(\omega t + \phi),$$

$$y(t) = A \sin(\omega t + \phi).$$

Squaring and adding,

$$x^2 + y^2 = A^2 \cos^2(\omega t + \phi) + A^2 \sin^2(\omega t + \phi).$$

Hence,

$$x^2 + y^2 = A^2.$$

Therefore, the point follows a circle of radius  $A$  centered at the origin:

$$\boxed{x^2 + y^2 = A^2.}$$

### (c) First and second derivatives

Differentiating,

$$\dot{z}(t) = i\omega A e^{i(\omega t + \phi)}.$$

Therefore,

$$\boxed{\dot{z}(t) = i\omega z(t).}$$

Differentiating again,

$$\ddot{z}(t) = (i\omega)^2 A e^{i(\omega t + \phi)}.$$

Since

$$i^2 = -1,$$

we have

$$\boxed{\ddot{z}(t) = -\omega^2 z(t).}$$

#### (d) Differential equation

From the previous result,

$$\ddot{z} = -\omega^2 z.$$

Therefore,

$$\boxed{\ddot{z} + \omega^2 z = 0.}$$

Thus the complex exponential automatically satisfies the differential equation of simple harmonic motion.

#### (e) Real and imaginary parts

Since

$$z(t) = x(t) + iy(t),$$

we have

$$\ddot{z} + \omega^2 z = (\ddot{x} + \omega^2 x) + i(\ddot{y} + \omega^2 y).$$

For this expression to vanish, both the real and imaginary parts must vanish:

$$\boxed{\ddot{x} + \omega^2 x = 0}$$

and

$$\boxed{\ddot{y} + \omega^2 y = 0.}$$

Thus both the real and imaginary components independently describe simple harmonic motion.

#### (f) Why use complex exponentials?

The complex exponential provides a compact representation of two real oscillatory functions:

$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t).$$

Differentiation becomes particularly simple:

$$\frac{d}{dt}e^{i\omega t} = i\omega e^{i\omega t}.$$

Similarly,

$$\frac{d^2}{dt^2}e^{i\omega t} = -\omega^2 e^{i\omega t}.$$

Thus, instead of carrying separate sine and cosine functions through every calculation, we can perform the calculation using a single complex exponential and take the real or imaginary part at the end.

The complex representation therefore provides a convenient mathematical tool for analyzing oscillatory motion, while the underlying physical motion can remain completely real.